Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

Level 1

Exercise 1

Based on the attached documents (estructura_dades and dades_introduir), import the two tables. Show the main features of the schema created and explain the different tables and variables that exist. Be sure to include a diagram that illustrates the relationship between the different tables and variables.

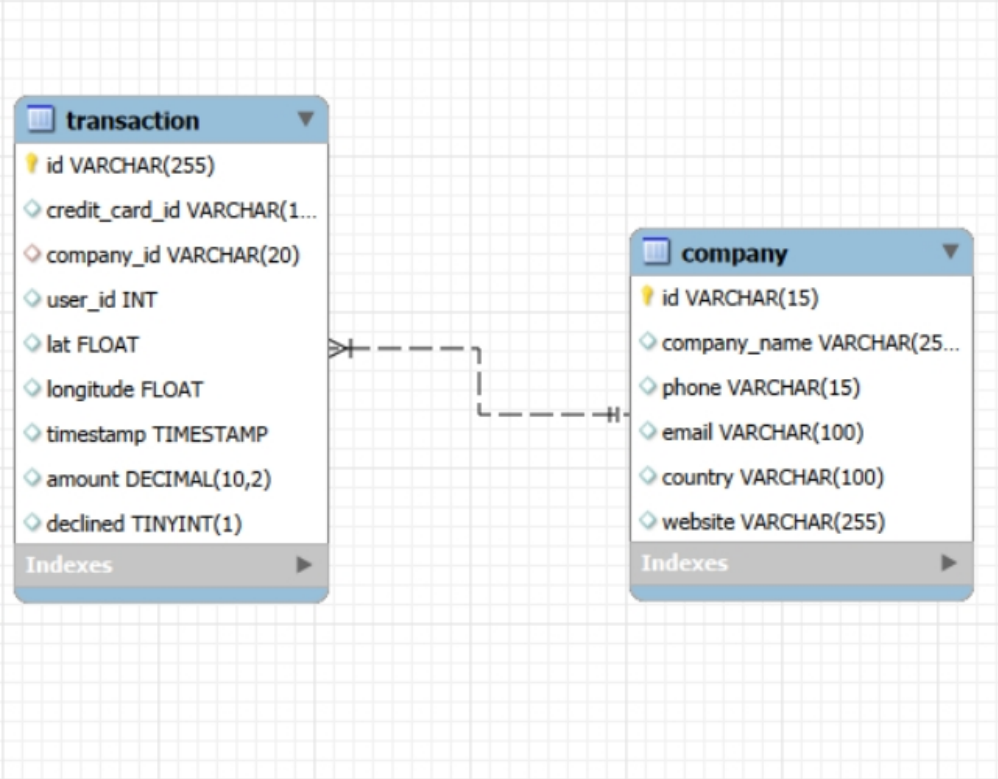


Fig 1. *ER diagram of the transactions database*

Figure 1 shows the ER diagram for the "transactions" database. This database includes two tables, `company` (dimensional table) and `transaction` (fact table), related via a 1:N granularity by the primary key "id" in the `company` table to the foreign key "company_id" in the `transaction` table (which has its own primary key: "id"). The ER diagram also illustrates the column names and their corresponding data types for each table.

Also, on the query tab, it is possible to fetch general information about database tables. Some examples are Figure 2 (using SHOW), and Figure 3 and Figure 4 (using DESCRIBE).

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

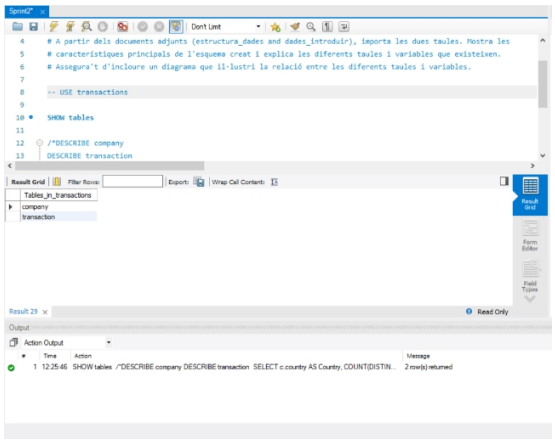


Fig 2. *Using SHOW to demonstrate all the tables of the database*

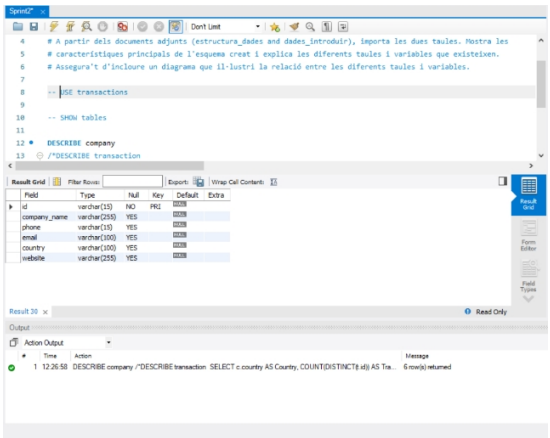


Fig 3. *Using DESCRIBE to have a general understanding of the* `company` *table*

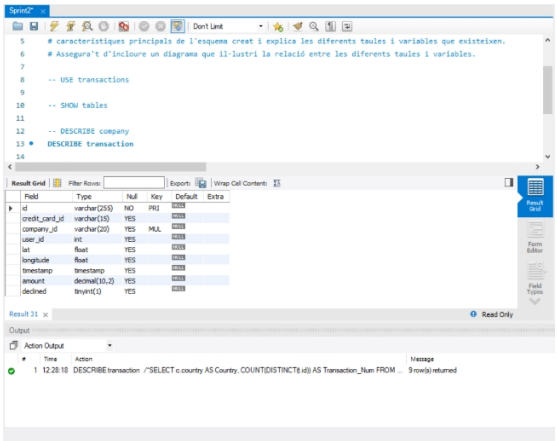


Fig 4. *Using DESCRIBE to have a general understanding of the* `transaction` *table*

Figure 5 is an example of going into more detail and learning more about the database's specifications, as are Figure 6 and Figure 7, which elaborate on the tables of our database.

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

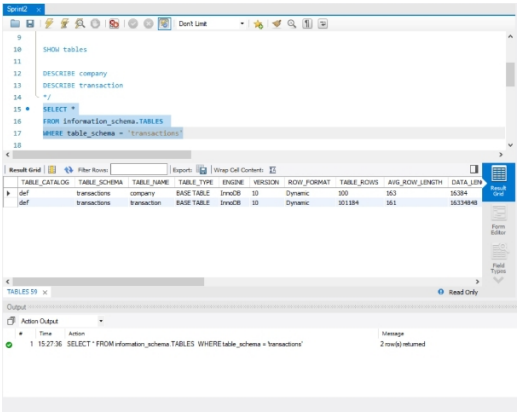


Fig 5. *Specifications of the* `transactions` *database*

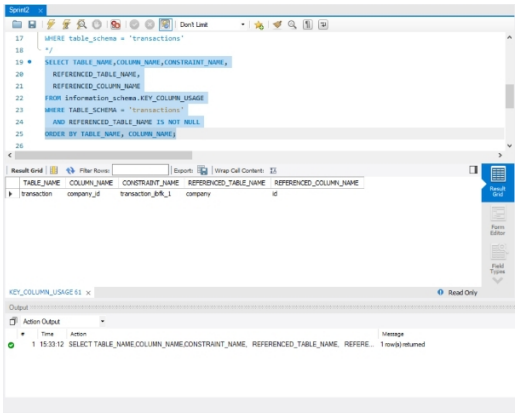


Fig 6. *Detailed relationship between the two tables of our database*

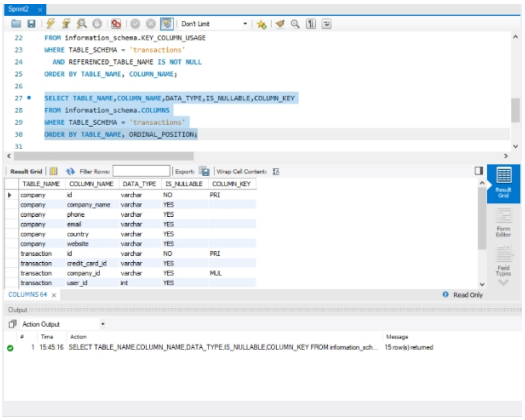


Fig 7. *Table-oriented details altogether*

In the end, Figure 8 and Figure 9 give us a general overview (e.g., number of rows, column names, etc.) of the `company` table and the `transaction` table, respectively.

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

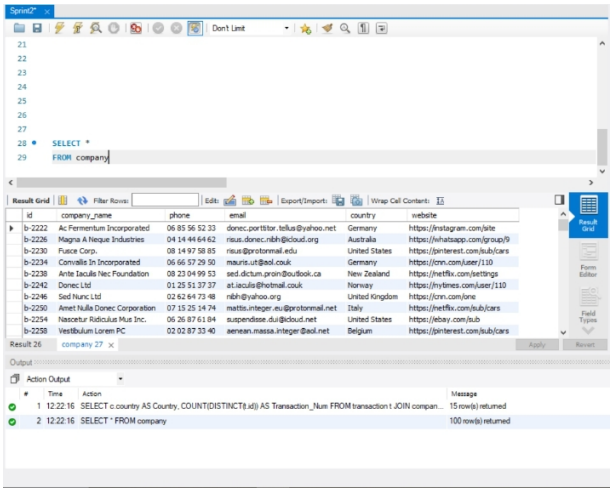


Fig 8. *company table*

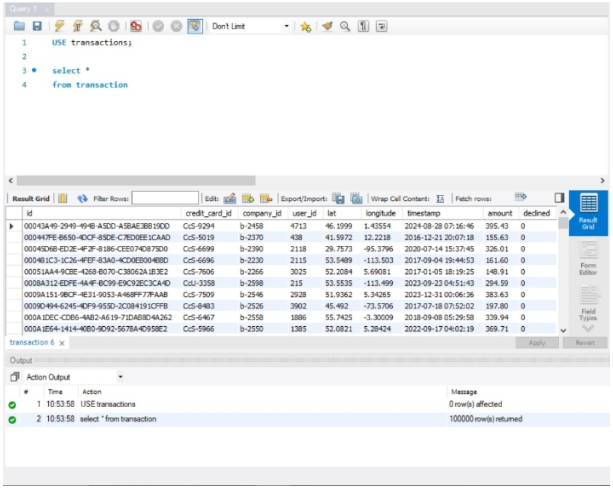


Fig 9. *transaction table*

Exercise 2

Using JOIN you will perform the following queries:

- List of countries that are generating sales.
- From how many countries are sales generated?
- Identify the company with the highest average sales.

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

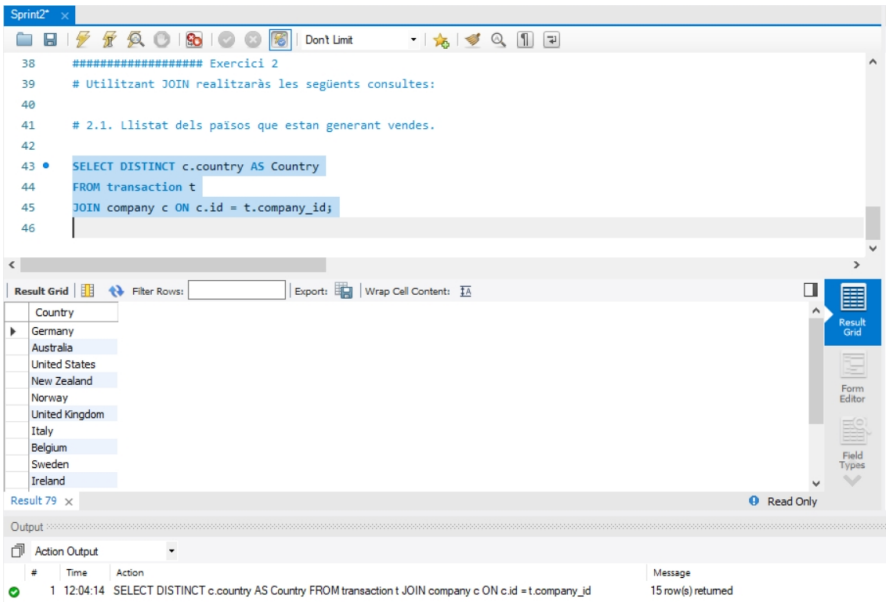


Fig 10. *List of countries that are generating sales.*

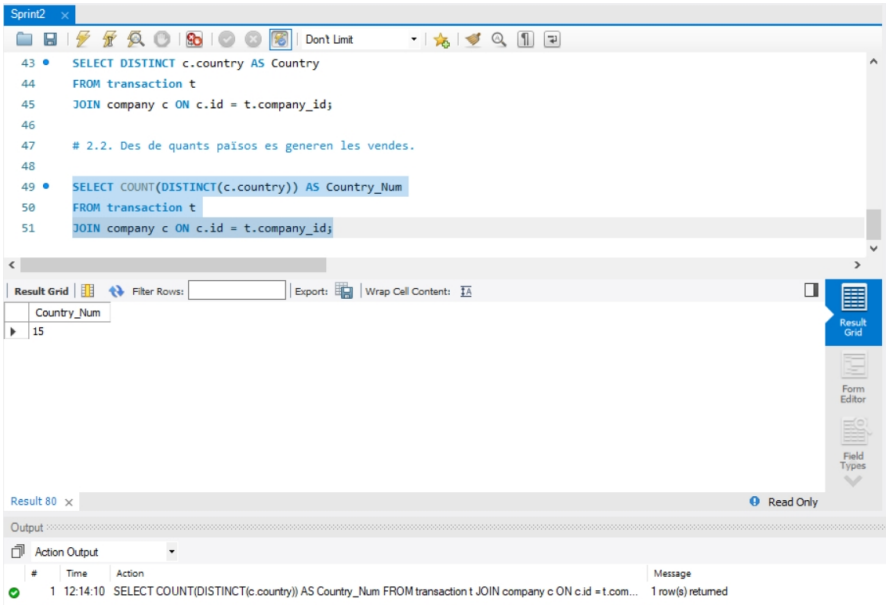


Fig 11. *The number of countries from which sales are generated.*

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

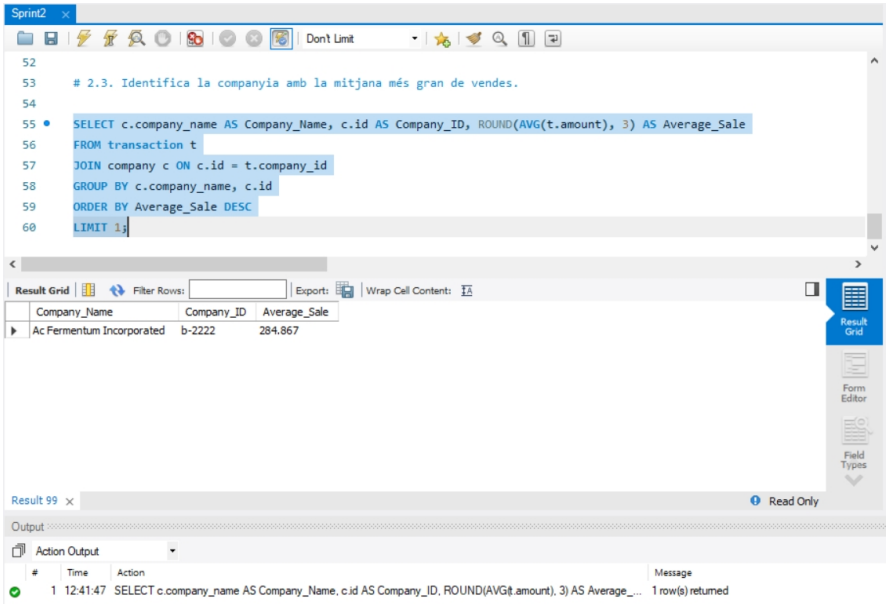


Fig 12. *The company with the highest average sales.*

Exercise 3

Using only subqueries (without using JOIN):

- It shows all transactions made by companies in Germany.
- Lists companies that have made transactions for an amount higher than the average of all transactions.
- They will eliminate from the system companies that do not have registered transactions, provide a list of these companies.

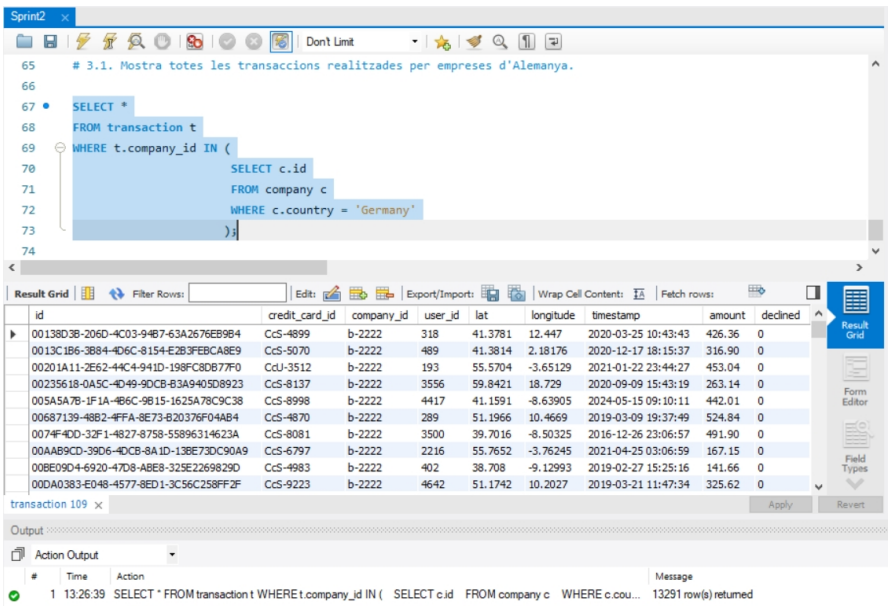


Fig 13. *All transactions made by companies in Germany.*

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

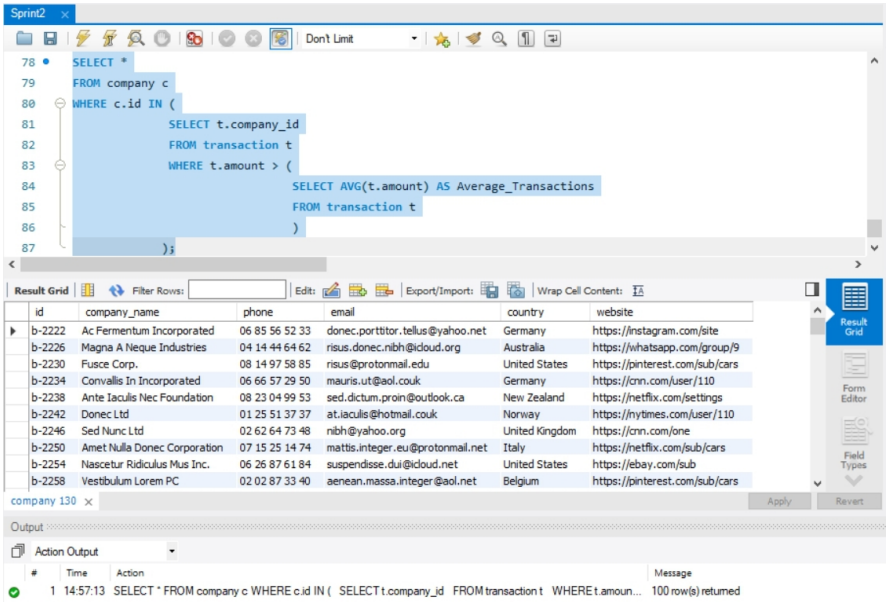


Fig 14. *Companies that have made transactions for an amount higher than the average of all transactions.*

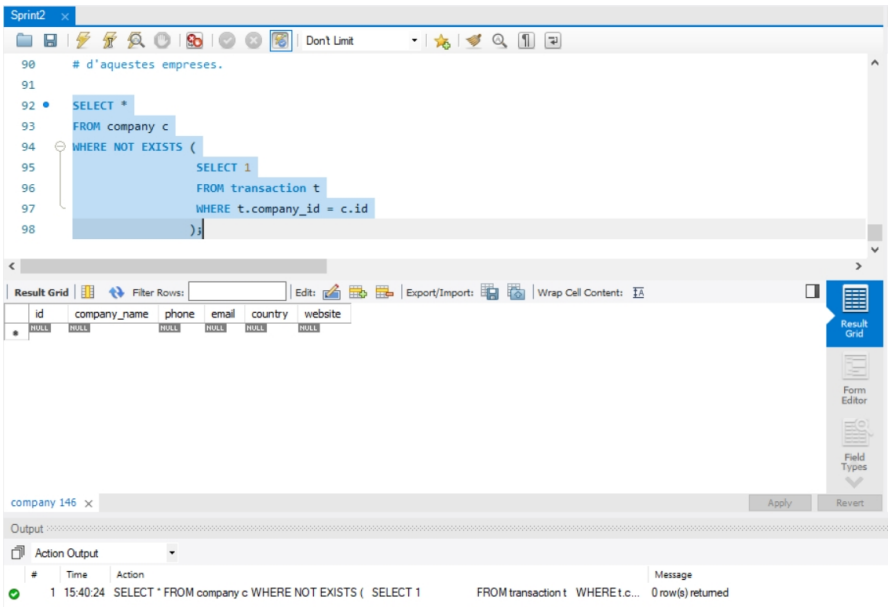


Fig 15. *Companies without registered transactions*

Level 2

Exercise 1

Identifies the five days that generated the highest amount of revenue for your business from sales. It shows the date of each transaction along with the total sales.

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

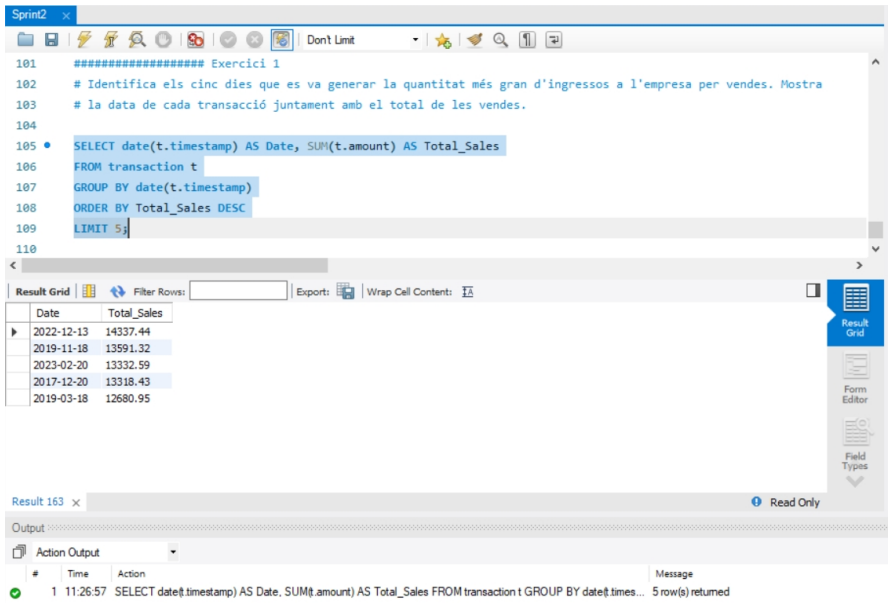


Fig. 16. *Five days that generated the highest amount of revenue (first approach based on the total sales)*

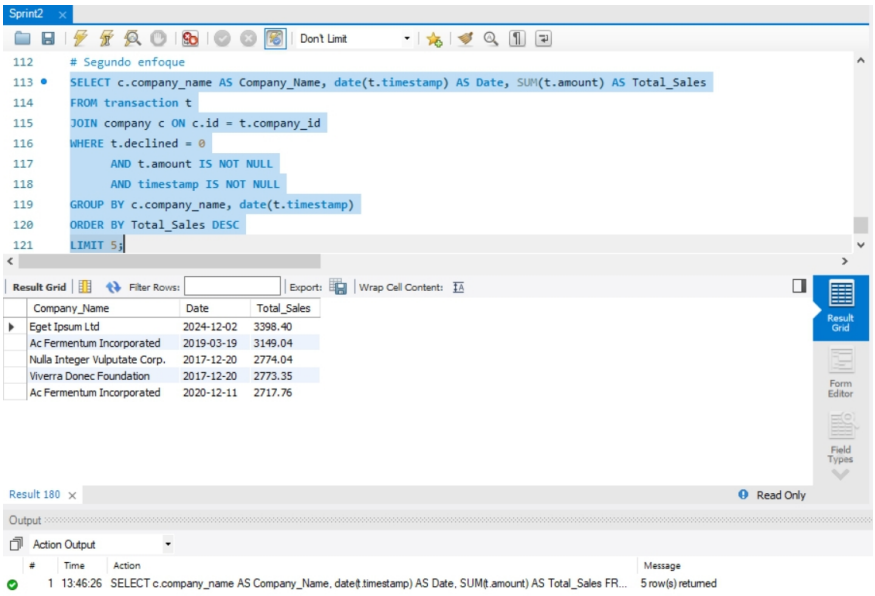


Fig. 17. *Five days that generated the highest amount of revenue (second approach based on each company's total sales)*

Exercise 2

What are the average sales by country? Presents the results sorted from highest to lowest average.

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

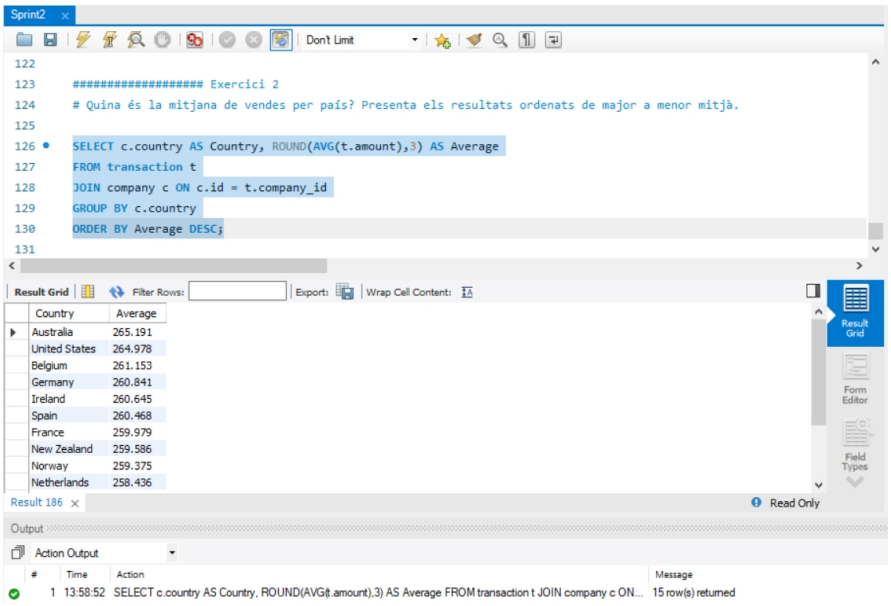


Fig 18. *Sorted average sales by country*

Exercise 3

In your company, a new project is being considered to launch some advertising campaigns to compete with the company "Non Institute". To do this, you are asked for a list of all the transactions carried out by companies that are located in the same country as this company.

- Show the list applying JOIN and subqueries.
- Show the list applying only subqueries.

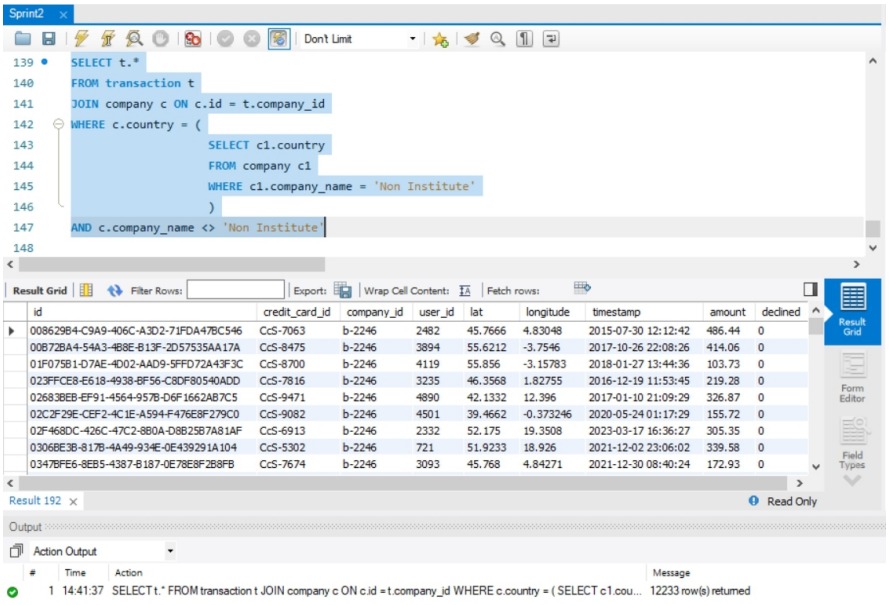


Fig 19. *All the transactions carried out by companies that are located in the UK except one, applying JOIN and subqueries*

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

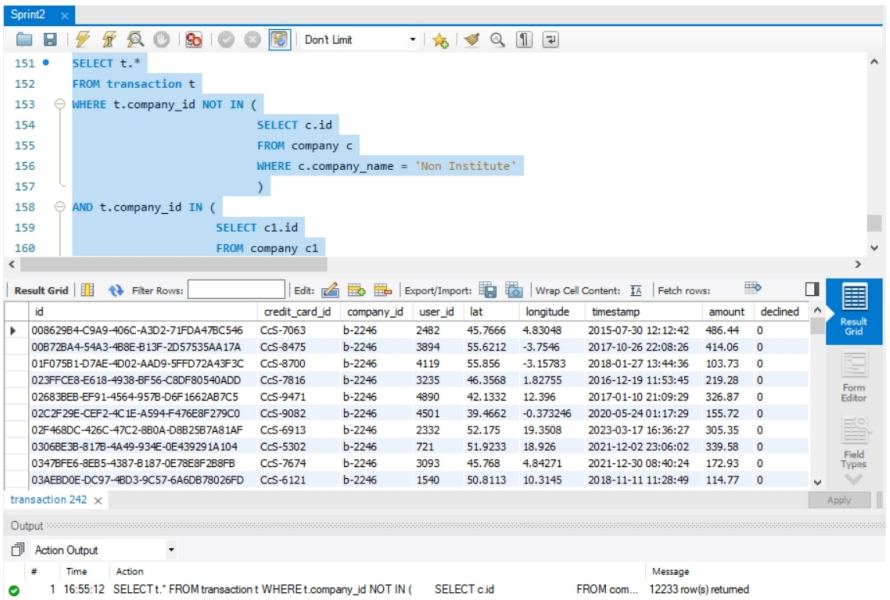


Fig 20. *All the transactions carried out by companies that are located in the UK except one, applying only subqueries*

Level 3

Exercise 1

It presents the name, telephone number, country, date and amount of those companies that carried out transactions with a value between 350 and 400 euros and on one of these dates: April 29, 2015, July 20, 2018 and March 13, 2024. Sort the results from highest to lowest amount.

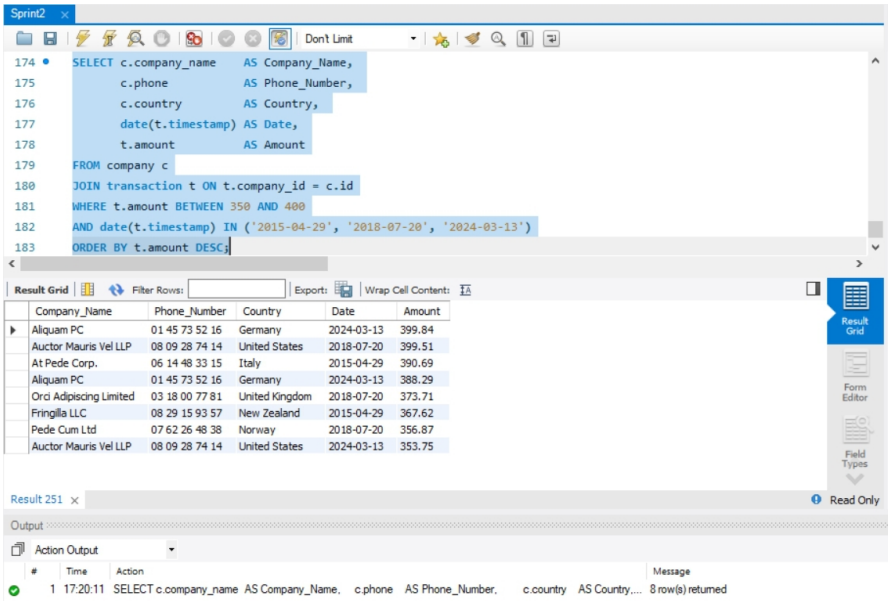


Fig 21. *Customized information for three specific dates*

Task S2.01. SQL Basics

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

Exercise 2

We need to optimize the allocation of resources and it will depend on the operational capacity required, so they ask you for information on the number of transactions that companies carry out, but the human resources department is demanding and wants a list of companies where you specify whether they have more than 400 transactions or less.

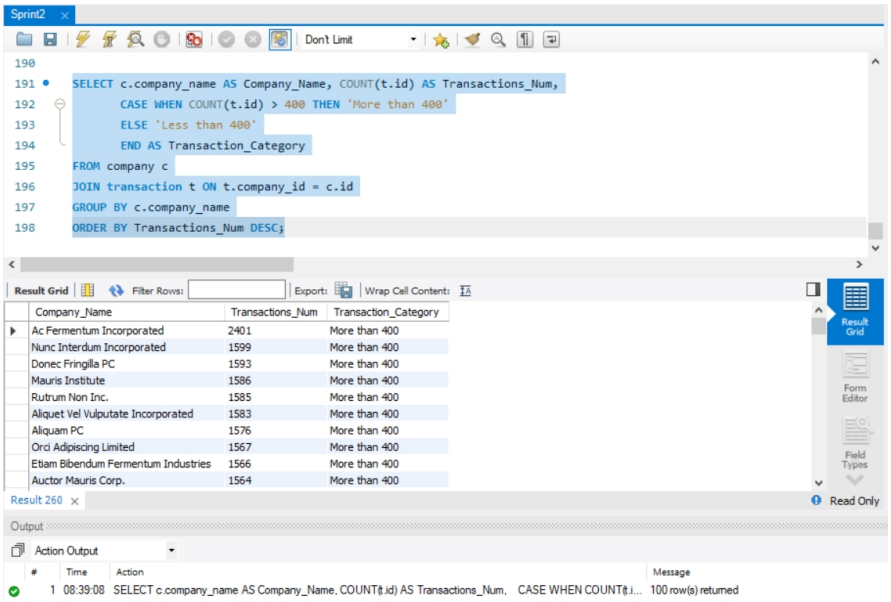


Fig 22. *Case-defined transactions*

It is advisable to include the following two lines of code to ensure there are no outliers or incorrect data in our results. However, since the final result is the same with and/or without them, the codes were not inserted:

t.declined = 0
t.amount IS NOT NULL